

H15 Semantic information through natural scrambles: which information keeps agents and the swarm viable

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

level: hypothesis · pipeline checks 85% · updated 2026-10-03

exploratory round 1 done (2026-10-03, non-holdout only). The pre-registered "memory is load-bearing" predictions failed: no detectable day-scale viability cost of losing up to ~80% of memory (P1) or of starting with an empty memory (P4); the rewrite control...

WHAT HAPPENED

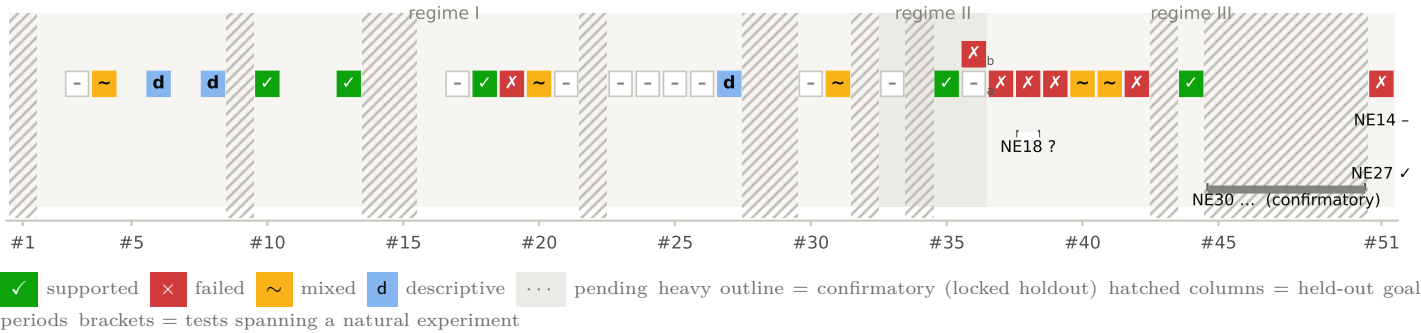
Natural scrambles in non-holdout data, per goal period, against a placebo-day null (a synthetic check on the real village skeleton: calibrated, power 17–41%). **Memory:** 12 losses of 50–80% of an agent’s memory and 20 newcomers with empty memories showed no day-scale cost on viability V^* : memory loss -0.55 SD $[-1.78, +0.68]$ (2 periods), newcomers $+0.15$ $[-0.18, +0.47]$ (more engaged early); the rewrite control was null. **Chat cuts:** only 3 usable events. **Context:** consolidations wipe the context window (forced every 41 turns) but keep memory. The pre-registered forced-vs-voluntary contrast came out wrong-signed (task-phase confound). Post hoc, writes drop 33–53% for ~10 turns after a forced wipe in 8/9 regime III periods, however much was saved to memory. D2.6 found no homeostatic V . Holdout not run.

PHYSICAL MODEL

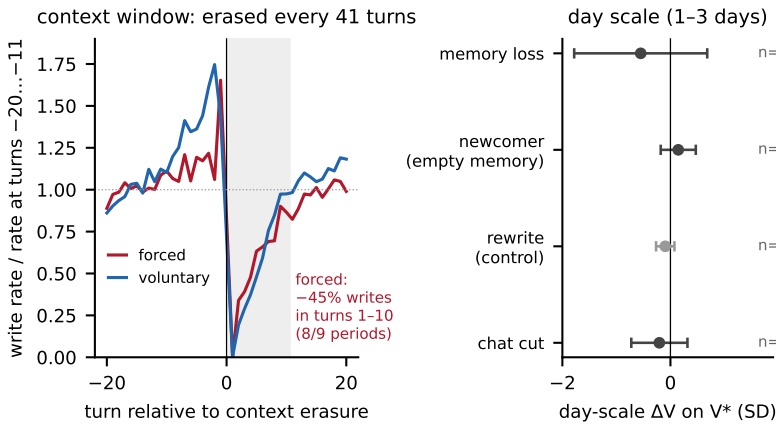
Model 04, Kolchinsky–Wolpert semantic information. X = one agent: memory M (day-scale store) and context window C (session-scale store, filled by chat and events); Y = the rest of the village. Scrambles erase M or C , or cut $Y \rightarrow X$; $\hat{p}$ = the scrambled joint. V = viability per agent-day (5 candidates) or per turn (writes); u = its same-day residual; δ = fraction removed; S, η = semantic information, its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$
$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$
$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$
$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t0-1} | u_{\text{pre}}])$$

GOAL PERIODS TESTED 6 supported, 7 failed, 5 mixed, 3 descriptive, 1 pending, 11 n/a



OBSERVABLES



WHAT THE FIGURE SHOWS

Left: write rate around consolidations, relative to turns $-20 \dots -11$ (regime III, pooled). A forced consolidation (41-turn cap, timing independent of task state) wipes the context: output falls $\sim 45\%$ for ten turns, then recovers (post-hoc). Voluntary ones dip alike after a write burst. Right: day-scale ΔV on V^* (meta over goal periods, 95% CI): memory loss, newcomers, rewrite control and chat cuts are all consistent with zero.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly confirms what people running LLM agents already suspect: the working context keeps an agent productive, and a long-term memory file is something it rebuilds from its environment within a day. The one usable number is the cost of a context reset, about 45% of writes for ~10 turns (about a tenth of output at a 41-turn cadence). The levers are a longer cap or a task-state summary carried across the reset; today’s memory write doesn’t recover the loss. That is scaffold engineering, not physics, and post-hoc until the holdout check (C3) runs. Kolchinsky–Wolpert adds vocabulary (stored vs. observed information) but here measured neither S nor η , nor a viability function.

Bottom line. Day-scale memory carries no detectable load-bearing information; the context window does (about -45% output for ten turns per forced wipe, post-hoc, awaiting the holdout).